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Taller 1
Sistemas Dinámicos y Control
Modelos Dinámicos

Tipo de trabajo: individual
Fecha de entrega: Día del primer examen
Forma de entrega: Archivos pdf cargados en Google Classroom

Problemas del libro "Feedback Control of Dynamic Systems – Eight Edition – Franklin, Powell, Emami-Naeini, Pearson".

Capítulo 2: Modelos dinámicos

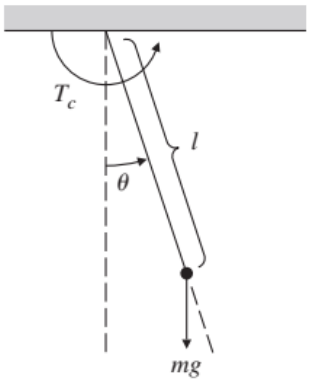
Movimiento traslacional

1. Problema 2.1 – página 99
2. Problema 2.5 – página 100: hace referencia al modelo de suspensión de un carro presentado en clase
3. Problema 2.8 – página 101

Movimiento rotacional

4. a. Write the equations of motion for the simple pendulum shown in Fig. 2.15, where all the mass is concentrated at the end point and there is a torque, Tc, applied at the pivot.
   b. Use Matlab to determine the time history of θ to a step input in Tc of 1 N·m. Assume l = 1 m, m = 1 kg, and g = 9.81 m/sec2.

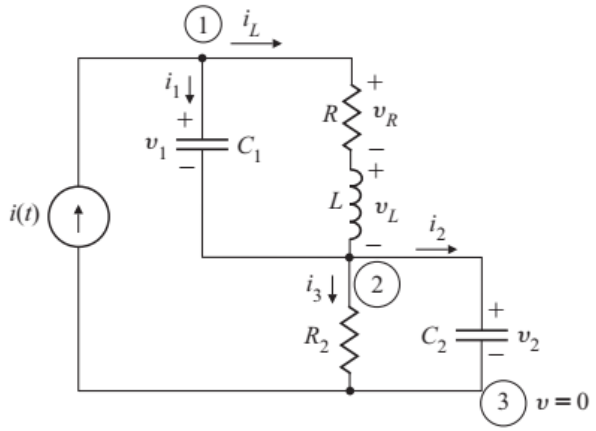
Figure 2.15
Pendulum



Circuitos eléctricos

5. Determine the differential equations for the circuit shown in Fig. 2.27. Choose the capacitor voltages and the inductor current as the unknowns.

Figure 2.27
Circuit for Example 2.10



6. Problema 2.17 – Página 105

Sistemas con fluidos
7. Problema 2.27 – Página 109
8. Problema 2.28 – Página 110

Capítulo 3: Respuesta dinámica

Transformada de Laplace
9. Problema 3.1 – página 187
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Efecto de la ubicación de los polos
13. Problema 3.25 – página 194

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14. Problema 3.28 – página 194
15. Problema 3.30 – página 195
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Estabilidad
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2.7 What is the major benefit of having zero input current to an operational amplifier?

2.8 Why is it important to have a small value for the armature resistance R_a of an electric motor?

2.9 What are the definition and units of the electric constant of a motor?

2.10 What are the definition and units of the torque constant of an electric motor?

2.11 Why do we approximate a physical model of the plant (which is *always* nonlinear) with a linear model?

△ **2.12** Give the relationships for the following:

(a) Heat flow across a substance

(b) Heat storage in a substance

△ **2.13** Name and give the equations for the three relationships governing fluid flow.

PROBLEMS

Problems for Section 2.1: Dynamics of Mechanical Systems

2.1 Write the differential equations for the mechanical systems shown in Fig. 2.43. For Fig. 2.43(a) and (b), state whether you think the system will eventually decay so it has no motion at all, given that there are nonzero

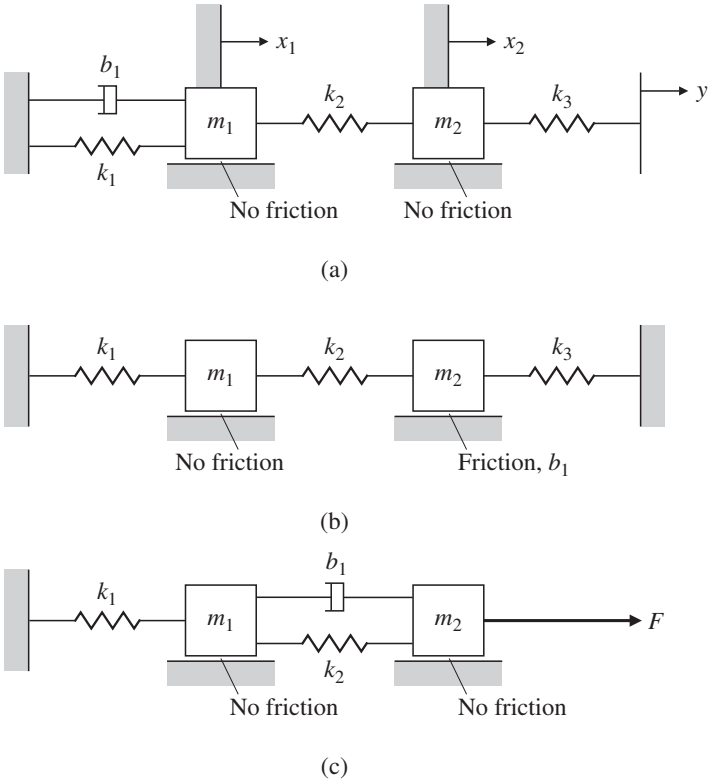
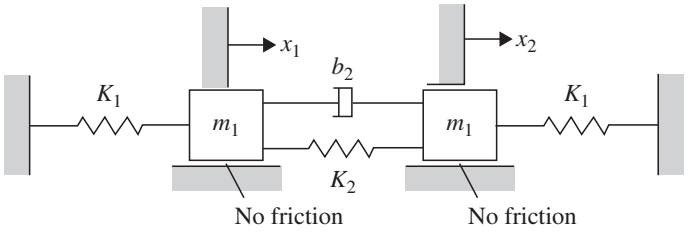


Figure 2.43
Mechanical systems

initial conditions for both masses and there is no input; give a reason for your answer. Also, for part (c), answer the question for $F = 0$.

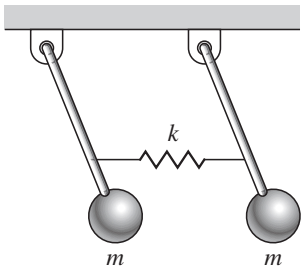
2.2 Write the differential equation for the mechanical system shown in Fig. 2.44. State whether you think the system will eventually decay so it has no motion at all, given that there are nonzero initial conditions for both masses, and give a reason for your answer.

Figure 2.44
Mechanical system for Problem 2.2



2.3 Write the equations of motion for the double-pendulum system shown in Fig. 2.45. Assume the displacement angles of the pendulums are small enough to ensure the spring is always horizontal. The pendulum rods are taken to be massless, of length l, and the springs are attached three-fourths of the way down.

Figure 2.45
Double pendulum



2.4 Write the equations of motion of a pendulum consisting of a thin, 2 kg stick of length l suspended from a pivot. How long should the rod be in order for the period to be exactly 1 sec? (The inertia I of a thin stick about an end point is $\frac{1}{3}ml^2$. Assume θ is small enough that $\sin\theta \cong \theta$.) Why do you think grandfather clocks are typically about 6 ft high?

2.5 For the car suspension discussed in Example 2.2, plot the position of the car and the wheel after the car hits a "unit bump"(that is, r is a unit step) using Matlab. Assume $m_1 = 10$ kg, $m_2 = 350$ kg, $K_w = 500{,}000$ N/m, and $K_s = 10{,}000$ N/m. Find the value of b that you would prefer if you were a passenger in the car.

2.6 For the quadcopter shown in Figs. 2.13 and 2.14:

(a) Determine the appropriate commands to rotor #s 1, 2, 3, & 4 so a pure vertical force will be applied to the quadcopter, that is, a force that will have no effect on pitch, roll, or yaw.

(b) Determine the transfer function between F_h, and altitude, h. That is, find $h(s)/F_h(s)$.

2.7 Automobile manufacturers are contemplating building active suspension systems. The simplest change is to make shock absorbers with a changeable damping, $b(u_1)$. It is also possible to make a device to be placed in parallel with the springs that has the ability to supply an equal force, u_2, in opposite directions on the wheel axle and the car body.

(a) Modify the equations of motion in Example 2.2 to include such control inputs.

(b) Is the resulting system linear?

(c) Is it possible to use the force u_2 to completely replace the springs and shock absorber? Is this a good idea?

2.8 In many mechanical positioning systems, there is flexibility between one part of the system and another. An example is shown in Fig. 2.7 where there is flexibility of the solar panels. Figure 2.46 depicts such a situation, where a force u is applied to the mass M and another mass m is connected to it. The coupling between the objects is often modeled by a spring constant k with a damping coefficient b, although the actual situation is usually much more complicated than this.

(a) Write the equations of motion governing this system.

(b) Find the transfer function between the control input u and the output y.

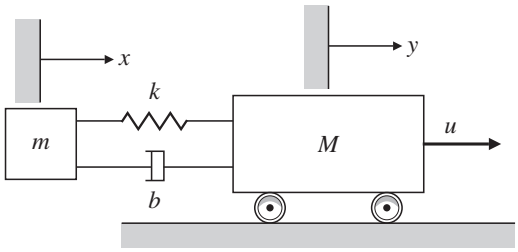


Figure 2.46
Schematic of a system with flexibility

2.9 Modify the equation of motion for the cruise control in Example 2.1, Eq. (2.4), so it has a control law; that is, let

$$u = K(v_r - v), \tag{2.111}$$

where

$$v_r = \text{reference speed}, \tag{2.112}$$

$$K = \text{constant}. \tag{2.113}$$

This is a "proportional"control law in which the difference between v_r and the actual speed is used as a signal to speed the engine up or slow it down. Revise the equations of motion with v_r as the input and v as the output and find the transfer function. Assume $m = 1500$ kg and $b = 70$ N·sec/m, and find the response for a unit step in v_r using Matlab. Using trial and error, find a value of K that you think would result in a control system in which the actual speed converges as quickly as possible to the reference speed with no objectionable behavior.

2.17 Write the dynamic equations and find the transfer functions for the circuits shown in Fig. 2.54.

(a) Passive lead circuit

(b) Active lead circuit

(c) Active lag circuit

(d) Passive notch circuit

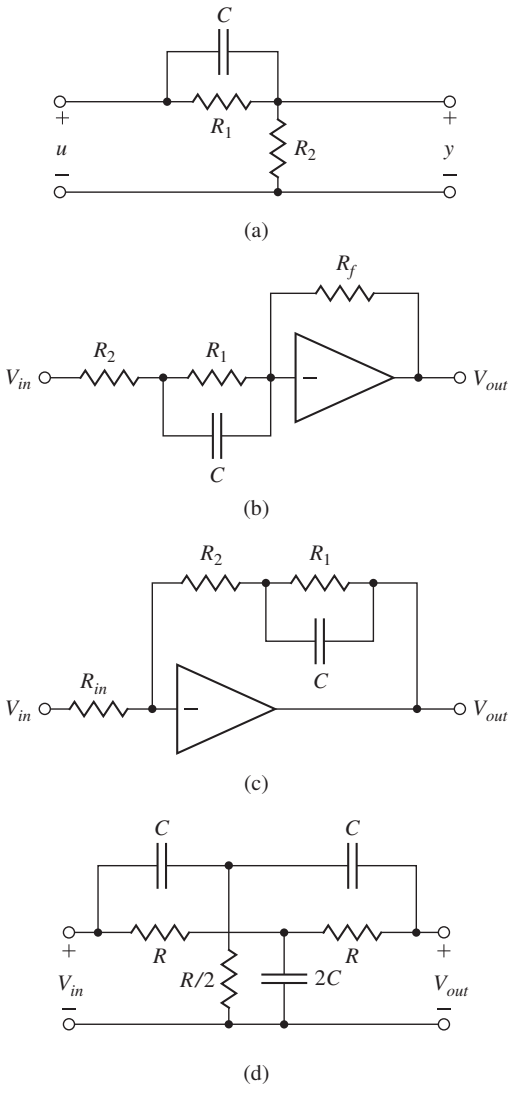


Figure 2.54
(a) Passive lead; (b) active lead; (c) active lag; and (d) passive notch circuits

2.18 The very flexible circuit shown in Fig. 2.55 is called a biquad because its transfer function can be made to be the ratio of two second-order or quadratic polynomials. By selecting different values for R_a, R_b, R_c, and R_d, the circuit can realize a low-pass, band-pass, high-pass, or band-reject (notch) filter.

(a) Show that if $R_a = R$ and $R_b = R_c = R_d = \infty$, the transfer function from V_{in} to V_{out} can be written as the low-pass filter

$$\frac{V_{out}}{V_{in}} = \frac{A}{\frac{s^2}{\omega_n^2} + 2\zeta\frac{s}{\omega_n} + 1}, \tag{2.114}$$

motion d is proportional to the difference between T_{act} and T_{amb} due to thermal expansion. Find the differential equations relating the height of the actuator d versus the applied voltage v_i.

2.26 An air conditioner supplies cold air at the same temperature to each room on the fourth floor of the high-rise building shown in Fig. 2.59(a). The floor plan is shown in Fig. 2.59(b). The cold airflow produces an equal amount of heat flow q out of each room. Write a set of differential equations governing the temperature in each room, where

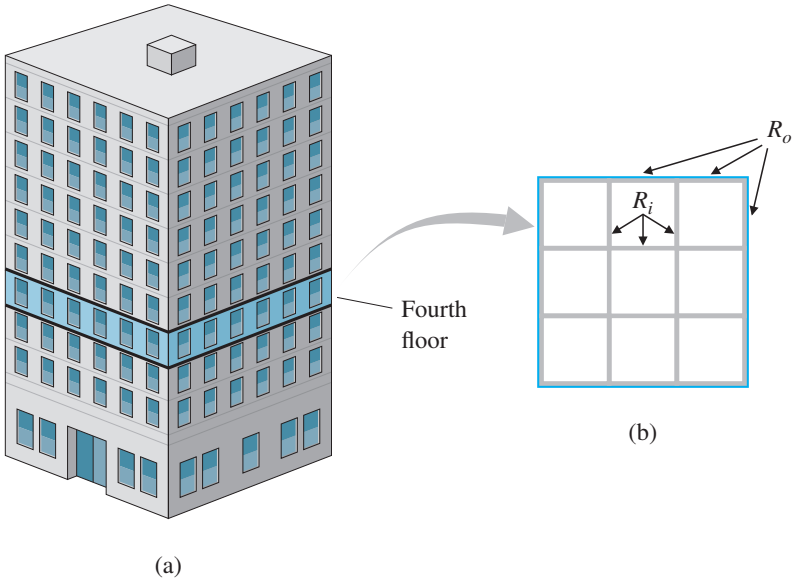


Figure 2.59
Building air-conditioning: (a) high-rise building; (b) floor plan of the fourth floor

T_o = temperature outside the building,

R_o = resistance to heat flow through the outer walls,

R_i = resistance to heat flow through the inner walls.

Assume (1) all rooms are perfect squares, (2) there is no heat flow through the floors or ceilings, and (3) the temperature in each room is uniform throughout the room. Take advantage of symmetry to reduce the number of differential equations to three.

2.27 For the two-tank fluid-flow system shown in Fig. 2.60, find the differential equations relating the flow into the first tank to the flow out of the second tank.

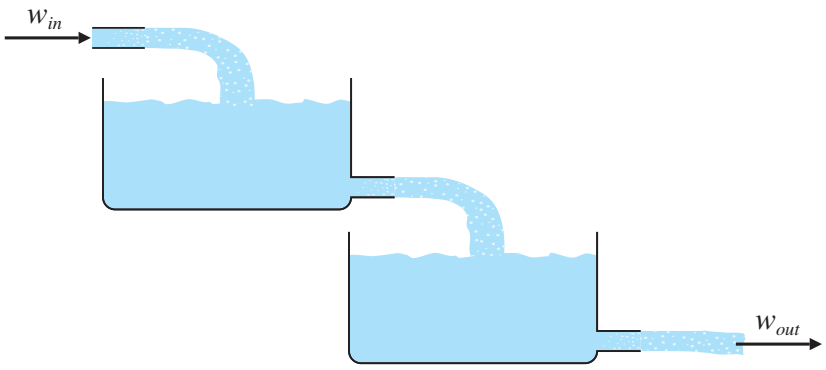


Figure 2.60
Two-tank fluid-flow system for Problem 2.27

2.28 A laboratory experiment in the flow of water through two tanks is sketched in Fig. 2.61. Assume Eq. (2.96) describes flow through the equal-sized holes at points A, B, or C.

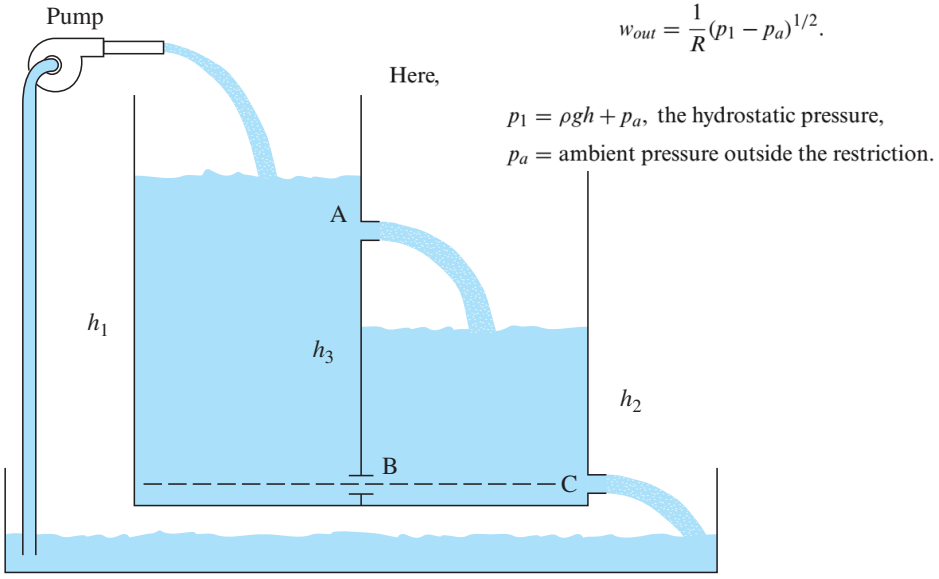


Figure 2.61
Two-tank fluid-flow system for Problem 2.28

$$w_{out} = \frac{1}{R}(p_1 - p_a)^{1/2}. \tag{2.96}$$

Here,

$p_1 = \rho g h + p_a$, the hydrostatic pressure,

p_a = ambient pressure outside the restriction.

(a) With holes at B and C, but none at A, write the equations of motion for this system in terms of h_1 and h_2. Assume when $h_2 = 15$ cm, the outflow is 200 g/min.

(b) At $h_1 = 30$ cm and $h_2 = 10$ cm, compute a linearized model and the transfer function from pump flow (in cubic-centimeters per minute) to h_2.

(c) Repeat parts (a) and (b) assuming hole B is closed and hole A is open. Assume $h_3 = 20$ cm, $h_1 > 20$ cm, and $h_2 < 20$ cm.

2.29 The equations for heating a house are given by Eqs. (2.81) and (2.82), and in a particular case can be written with time in *hours* as

$$C\frac{dT_h}{dt} = Ku - \frac{T_h - T_o}{R},$$

where

(a) C is the thermal capacity of the house, BTU/°F,

(b) T_h is the temperature in the house, °F,

(c) T_o is the temperature outside the house, °F,

(d) K is the heat rating of the furnace, $= 90,000$ BTU/h,

(e) R is the thermal resistance, °F per BTU/h,

(f) u is the furnace switch, $= 1$ if the furnace is on and $= 0$ if the furnace is off.

It is measured that, with the outside temperature at 32°F and the house at 60°F, the furnace raises the temperature 2°F in six min (0.1 h). With the furnace off, the house temperature falls 2°F in 40 min. What are the values of C and R for the house?

REVIEW QUESTIONS

3.1 What is the definition of a "transfer function"?

3.2 What are the properties of systems whose responses can be described by transfer functions?

3.3 What is the Laplace transform of $f(t-\lambda)1(t-\lambda)$ if the transform of $f(t)$ is $F(s)$?

3.4 State the Final Value Theorem (FVT).

3.5 What is the most common use of the FVT in control?

3.6 Given a second-order transfer function with damping ratio ζ and natural frequency ω_n, what is the estimate of the step response rise time? What is the estimate of the percent overshoot in the step response? What is the estimate of the settling time?

3.7 What is the major effect of a zero in the left half-plane on the second-order step response?

3.8 What is the most noticeable effect of a zero in the right half-plane on the step response of the second-order system?

3.9 What is the main effect of an extra real pole on the second-order step response?

3.10 Why is stability an important consideration in control system design?

3.11 What is the main use of Routh's criterion?

3.12 Under what conditions might it be important to know how to estimate a transfer function from experimental data?

PROBLEMS

Problems for Section 3.1: Review of Laplace Transforms

3.1 Show that, in a partial-fraction expansion, complex conjugate poles have coefficients that are also complex conjugates. (The result of this relationship is that whenever complex conjugate pairs of poles are present, only one of the coefficients needs to be computed.)

3.2 Find the Laplace transform of the following time functions:

(a) $f(t) = 0.5 + 2.5t$

(b) $f(t) = 1.5 + 9t + 0.3t^2 + \delta(t)$, where $\delta(t)$ is the unit impulse function

(c) $f(t) = 5.5e^{-t} + 3e^{-2t} + 2.5t^2e^{-3t}$

(d) $f(t) = (2t+1)^2$

(e) $f(t) = \cosh 0.2t$

3.3 Find the Laplace transform of the following time functions:

(a) $f(t) = 8\sin 0.75t$

(b) $f(t) = \cos 1.5t + 4\sin 1.5t + 1.7e^{-0.5t}\cos 1.5t$

(c) $f(t) = 0.4t^3 + 1.8e^t \sin 2.2t$

3.4 Find the Laplace transform of the following time functions:

(a) $f(t) = t\cos t$

(b) $f(t) = t\sin 0.9t$

(c) $f(t) = te^{-0.1t} + 2t\cos 0.6t$

(d) $f(t) = 3t^2 - 2t\sin 1.3t + 5t\cos 7.2t$

(e) $f(t) = t^2\cos t + t^2\sin 0.9t$

3.5 Find the Laplace transform of the following time functions (* denotes convolution):

(a) $f(t) = \sin 5t\cos t$

(b) $f(t) = 2 + 3\sin^2 4t + 5\cos^2 t$

(c) $f(t) = (\sin t)/t$

(d) $f(t) = \sin t * \cos 2t$

(e) $f(t) = \int_0^t \sin(t-\tau)\sin\tau\, d\tau$

3.6 Given the Laplace transform of $f(t)$ is $F(s)$, find the Laplace transform of the following:

(a) $g(t) = f(t)\cos t$

(b) $g(t) = \int_0^t \int_0^{t_1} f(\tau)d\tau dt_1$

3.7 Find the time function corresponding to each of the following Laplace transforms using partial-fraction expansions:

(a) $F(s) = \frac{5}{s(s+7)}$

(b) $F(s) = \frac{6}{s(s+1)(s+2)}$

(c) $F(s) = \frac{8s+2}{s^2+s+20}$

(d) $F(s) = \frac{5s+3}{(s+1)(s^2+2s+30)}$

(e) $F(s) = \frac{s+4}{s^2+2}$

(f) $F(s) = \frac{s+1}{s(s^2+4)}$

(g) $F(s) = \frac{s+9}{s^2(s+1)}$

(h) $F(s) = \frac{5}{s^5}$

(i) $F(s) = \frac{18}{s^4+9}$

(j) $F(s) = \frac{e^{-3}}{s^3}$

3.8 Find the time function corresponding to each of the following Laplace transforms:

(a) $F(s) = \frac{1}{s(s+2)^2}$

(b) $F(s) = \frac{s^2+s+1}{s^3-1}$

(c) $F(s) = \frac{2(s^2+s+1)}{s(s+1)^2}$

(d) $F(s) = \frac{s^3+2s+4}{s^4-16}$

(e) $F(s) = \frac{2(s+2)(s+5)^2}{(s+1)(s^2+4)^2}$

(f) $F(s) = \frac{s^2-1}{(s^2+1)^2}$

(g) $F(s) = \tan^{-1}\left(\frac{1}{s}\right)$

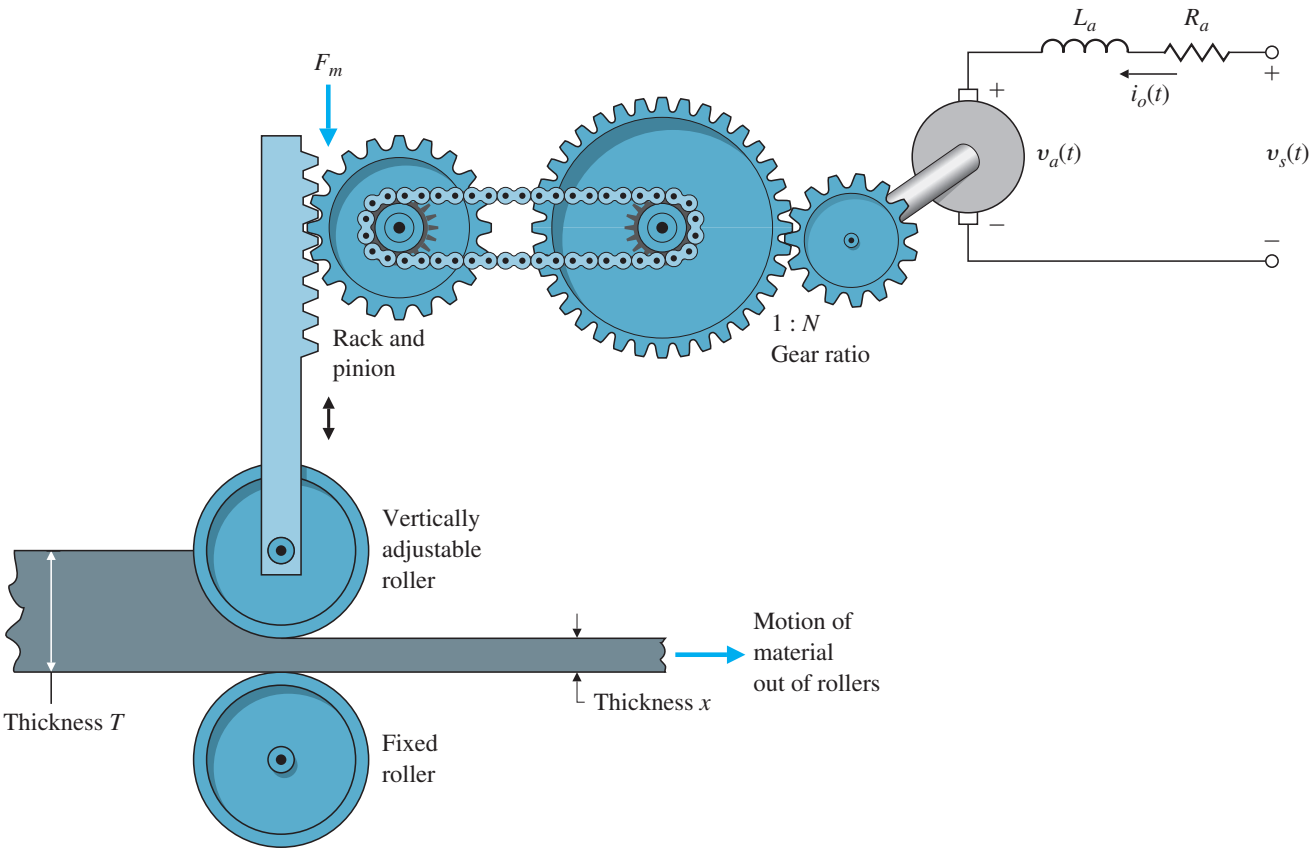
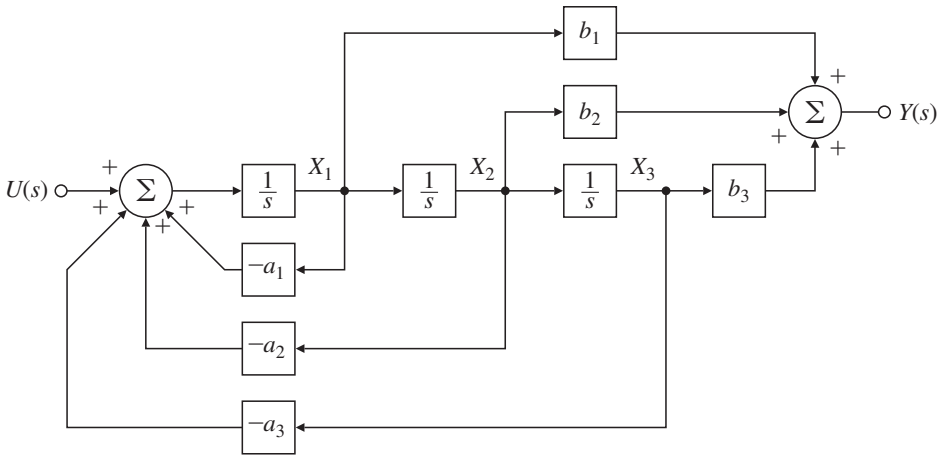


Figure 3.48
Continuous rolling mill

Problems for Section 3.2: System Modeling Diagrams

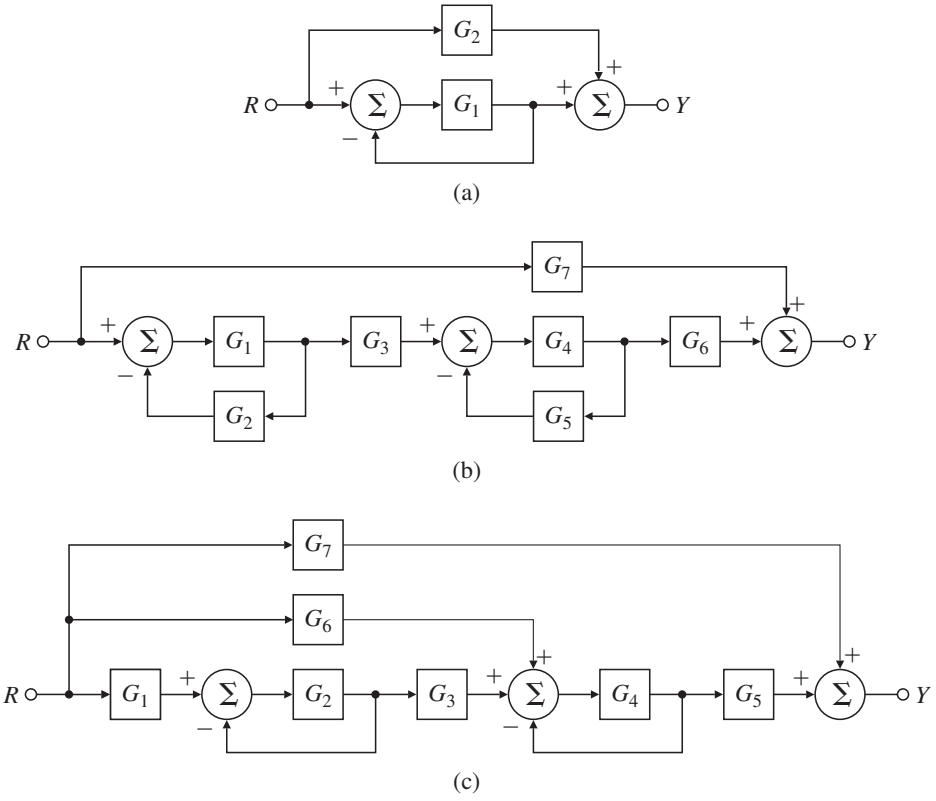
3.18 Consider the block diagram shown in Fig. 3.49. Note a_i and b_i are constants. Compute the transfer function for this system. This special structure is called the "control canonical form", and will be discussed further in Chapter 7.

Figure 3.49
Block diagram for Problem 3.18



3.19 Find the transfer functions for the block diagrams in Fig. 3.50.

Figure 3.50
Block diagrams for Problem 3.19



3.20 Find the transfer functions for the block diagrams in Fig. 3.51, using the ideas of block-diagram simplification. The special structure in Fig. 3.51(b) is called the "observer canonical form", and will be discussed in Chapter 7.

3.21 Use block-diagram algebra to determine the transfer function between $R(s)$ and $Y(s)$ in Fig. 3.52.

△ **3.22** Find the transfer functions for the block diagrams in Fig. 3.51, using Mason's rule.

△ **3.23** Use Mason's rule to determine the transfer function between $R(s)$ and $Y(s)$ in Fig. 3.52.

Problems for Section 3.3: Effect of Pole Locations

3.24 For the electric circuit shown in Fig. 3.53, find the following:

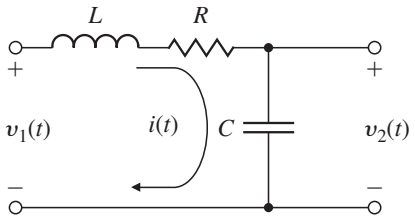
(a) The time-domain equation relating $i(t)$ and $v_1(t)$;

(b) The time-domain equation relating $i(t)$ and $v_2(t)$;

(c) Assuming all initial conditions are zero, the transfer function $\frac{v_2(s)}{v_1(s)}$ and the damping ratio ζ and undamped natural frequency ω_n of the system;

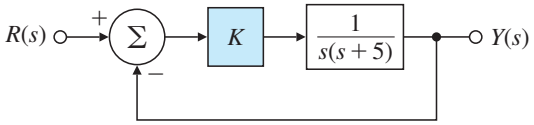
(d) The range of C values that will result in $v_2(t)$ having an overshoot of no more than 20%, assuming $v_1(t)$ is a unit step, $L = 1.5$ mH, and $R = 5\Omega$.

Figure 3.53
Circuit for Problem 3.24



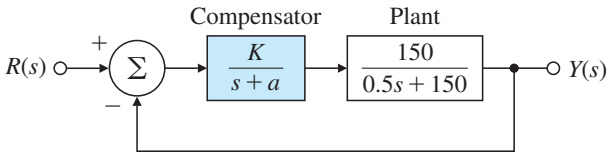
3.25 For the unit feedback system shown in Fig. 3.54, specify the gain K of the proportional controller so that the output $y(t)$ has an overshoot of no more than 12% in response to a unit step.

Figure 3.54
Unity feedback system for Problem 3.25



3.26 For the unity feedback system shown in Fig. 3.55, specify the gain and pole location of the compensator so that the overall closed-loop response to a unit-step input has an overshoot of no more than 18% and a 1% settling time of no more than 0.01 sec. Verify your design using Matlab.

Figure 3.55
Unity feedback system for Problem 3.26



Problems for Section 3.4: Time-Domain Specification

3.27 Suppose you desire the peak time of a given second-order system to be less than t_p'. Draw the region in the s-plane that corresponds to values of the poles that meet the specification $t_p < t_p'$.

3.28 A certain servomechanism system has dynamics dominated by a pair of complex poles and no finite zeros. The time-domain specifications on the rise time (t_r), percent overshoot (M_p), and settling time (t_s) are given by:

$$t_r \leq 0.6\,\text{sec},$$
$$M_p \leq 17\%,$$
$$t_s \leq 9.2\,\text{sec}.$$

(a) Sketch the region in the s-plane where the poles could be placed so that the system will meet *all* three specifications.

(b) Indicate on your sketch the specific locations (denoted by $\times$) that will have the smallest rise time and also meet the settling time specification *exactly*.

3.29 A feedback system has the following response specifications:

- Percent overshoot $M_p \leq 16\%$
- Settling time $t_s \leq 6.9$ sec
- Rise time $t_r \leq 1.8$ sec

(a) Sketch the region of acceptable closed-loop poles in the s-plane for the system, assuming the transfer function can be approximated as simple second order.

(b) What is the expected overshoot if the rise time and settling time specifications are met *exactly*?

3.30 Suppose you are to design a unity feedback controller for a first-order plant depicted in Fig. 3.56. (As you will learn in Chapter 4, the configuration shown is referred to as a proportional–integral controller.) You are to design the controller so that the closed-loop poles lie within the shaded regions shown in Fig. 3.57.

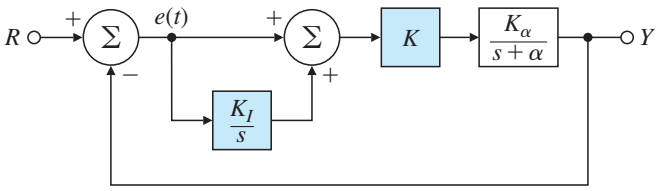


Figure 3.56
Unity feedback system for Problem 3.30

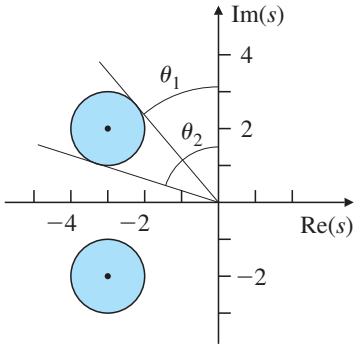


Figure 3.57
Desired closed-loop pole locations for Problem 3.30

(a) What values of ω_n and ζ correspond to the shaded regions in Fig. 3.57? (A simple estimate from the figure is sufficient.)

(b) Let $K_\alpha = \alpha = 2$. Find values for K and K_I so the poles of the closed-loop system lie within the shaded regions.

(c) Prove that no matter what the values of K_α and α are, the controller provides enough flexibility to place the poles anywhere in the complex (left-half) plane.

3.31 The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s(s+2)}.$$

The desired system response to a step input is specified as peak time $t_p = 1$ sec and overshoot $M_p = 5\%$.

3.45 Six transfer functions with unity DC gain are given below.

(a) Which transfer function(s) will meet an overshoot specification of $M_p \leq 17\%$?

(b) Which transfer function(s) will meet a rise time specification of $t_r \leq 0.3$ sec?

(c) Which transfer function(s) will meet a settling time specification of $t_s \leq 1.3$ sec?

$$G_1(s) = \frac{53.5}{(s^2 + 7.31s + 53.5)},$$

$$G_2(s) = \frac{313}{(s + 5.85)(s^2 + 7.31s + 53.5)},$$

$$G_3(s) = \frac{313}{0.5319(s + 11)(s^2 + 7.31s + 53.5)},$$

$$G_4(s) = \frac{5.9(s + 9.1)}{(s^2 + 7.31s + 53.5)},$$

$$G_5(s) = \frac{9.8(s^2 + 8s + 60)}{(s + 11)(s^2 + 7.31s + 53.5)},$$

$$G_6(s) = \frac{1.78(s^2 + 8s + 60)}{(s + 2)(s^2 + 7.31s + 53.5)}.$$

3.46 Consider the following two nonminimum-phase systems:

$$G_1(s) = -\frac{2(s - 1)}{(s + 1)(s + 2)}, \tag{3.98}$$

$$G_2(s) = \frac{3(s - 1)(s - 2)}{(s + 1)(s + 2)(s + 3)}. \tag{3.99}$$

(a) Sketch the unit-step responses for $G_1(s)$ and $G_2(s)$, paying close attention to the transient part of the response.

(b) Explain the difference in the behavior of the two responses as it relates to the zero locations.

(c) Consider a stable, strictly proper system (that is, m zeros and n poles, where $m < n$). Let $y(t)$ denote the step response of the system. The step response is said to have an undershoot if it initially starts off in the "wrong" direction. Prove that a stable, strictly proper system has an undershoot if and only if its transfer function has an *odd* number of *real* RHP zeros.

3.47 Find the relationships for the impulse response and the step response corresponding to Eq. (3.65) for the cases where

(a) the roots are repeated.

(b) the roots are both real. Express your answers in terms of hyperbolic functions (sinh, cosh) to best show the properties of the system response.

(c) the value of the damping coefficient, ζ, is negative.

3.48 Consider the following second-order system with an extra pole:

$$H(s) = \frac{\omega_n^2 p}{(s + p)(s^2 + 2\zeta\omega_n s + \omega_n^2)}.$$

Using Matlab, find a value of K that will provide an overshoot of less than 10% and a rise time faster than 0.5 sec for a unit-step change in θ_r. After examining the step response of the system for various values of K, comment on the difficulty associated with meeting rise time and overshoot specifications for complicated systems.

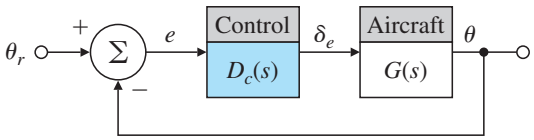


Figure 3.64
Block diagram of autopilot for Problem 3.50

Problems for Section 3.6: Stability

3.51 A measure of the degree of instability in an unstable aircraft response is the amount of time it takes for the *amplitude* of the time response to double (see Fig. 3.65), given some nonzero initial condition.

(a) For a first-order system, show that the **time to double** is

$$\tau_2 = \frac{\ln 2}{p}$$

where p is the pole location in the RHP.

(b) For a second-order system (with two complex poles in the RHP), show that

$$\tau_2 = \frac{\ln 2}{-\zeta\omega_n}.$$

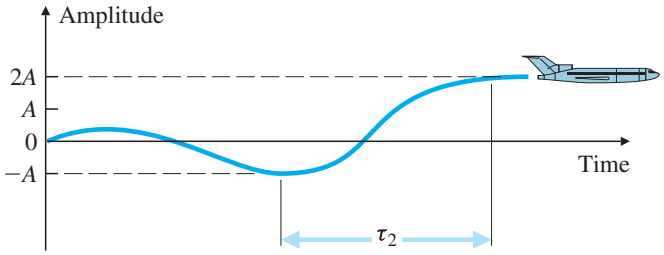


Figure 3.65
Time to double

3.52 Suppose that unity feedback is to be applied around the listed open-loop systems. Use Routh's stability criterion to determine whether the resulting closed-loop systems will be stable.

(a) $K(s)G(s) = \dfrac{5(s+5)}{(s+1)(s^3+2s+5)}$

(b) $K(s)G(s) = \dfrac{0.2(s^2+0.95s+0.11)}{s(s^2+0.36s+0.72)}$

(c) $K(s)G(s) = \dfrac{(s^3+15.5s^2+12.2s+100)}{(s+1)^2(47.7s^3+23.4s^2+20.3s+1)}$

3.53 Use Routh's stability criterion to determine how many roots with positive real parts the following equations have:

(a) $s^4 + 5.2s^3 + 18.9s^2 + 43.2s + 45.4$

(b) $s^5 + 0.102s^4 + 1.123s^3 + 0.686s^2 + 0.154s + 2$

(c) $s^4 + 152s^3 + 12s^2 - 1932s - 4921$

(d) $99s^3 - s^2 - 6s - 7$

(e) $s^4 + 8s^2 + 36$

3.54 Find the range of K for which all the roots of the following polynomial are in the LHP:

$$s^5 + 5s^4 + 10s^3 + 10s^2 + 5s + K = 0.$$

Use Matlab to verify your answer by plotting the roots of the polynomial in the s-plane for various values of K.

3.55 The transfer function of a typical tape-drive system is given by

$$KG(s) = \frac{K(s+6)}{s\left[(s+0.7)(s+1.2)(s^2+0.8s+6)\right]},$$

where time is measured in milliseconds. Using Routh's stability criterion, determine the range of K for which this system is stable when the characteristic equation is $1 + KG(s) = 0$.

3.56 Consider the closed-loop magnetic levitation system shown in Fig. 3.66. Determine the conditions on the system parameters (a, K, z, p, K_o) to guarantee closed-loop system stability.

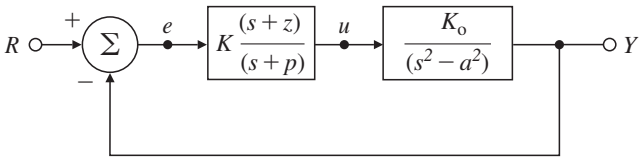


Figure 3.66
Magnetic levitation system for Problem 3.56

3.57 Consider the system shown in Fig. 3.67.

(a) Compute the closed-loop characteristic equation.

(b) For what values of (T, A) is the system stable? *Hint*: An approximate answer may be found using

$$e^{-Ts} \cong 1 - Ts,$$

or

$$e^{-Ts} \cong \frac{1 - \frac{T}{2}s}{1 + \frac{T}{2}s},$$

for the pure delay. As an alternative, you could use the computer Matlab (Simulink) to simulate the system or to find the roots of the system's characteristic equation for various values of T and A.

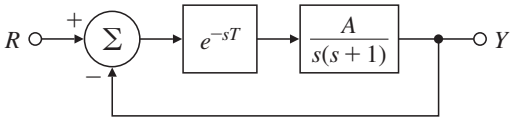


Figure 3.67
Control system for Problem 3.57